

N-gram language models

There are limitations of unigram language model

atriptothethemepark → a trip to the the me park

We all know that this is an unlikely segmentation.

Why do we know that?

Despite that "the" and "me" are very common words?

Unigram model: a model of word probabilities that does not take the word context into account.

atriptothethemepark $\rightarrow$ a trip to the the me park

$P(a) * P(trip) * P(to) * P(the) * P(the) * P(me) * P(park) >$

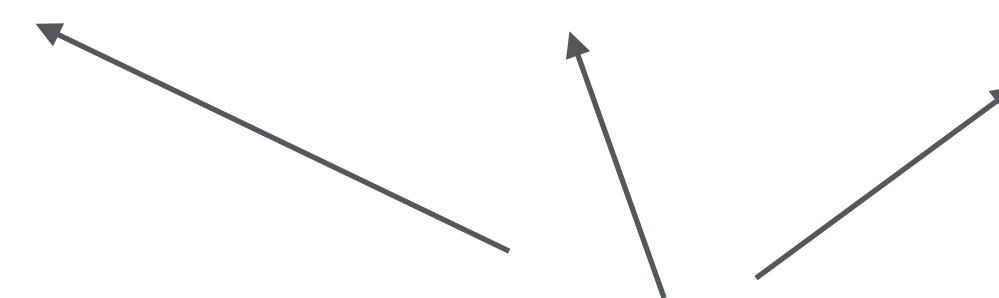
$P(a) * P(trip) * P(to) * P(the) * P(theme) * P(park)$

N-gram model: A model that uses probabilities for a word, given its preceding word(s).

$N = 2$:

E.g., the probability of the word "me", when "the" is the preceding word.

$P(a|PAD) * P(trip|a) * P(to|trip) * P(the|to) * P(the|the) * P(me|the) * P(park|me)$



Unlikely combinations

atriptothethemepark $\rightarrow$ a trip to the the me park OR a trip to the theme park

N-gram model: A model that uses probabilities for a word, given its preceding word(s).

$N = 2$:

$\mathbf{P(a|PAD)*P(trip|a)*P(to|trip)*P(the|to)*P(the|the)*P(me|the)*P(park|me)}$

Unlikely combinations

$\mathbf{P(a|PAD)*P(trip|a)*P(to|trip)*P(the|to)*P(theme|the)*P(park|theme)}$ $\leftarrow$ Winner

PAD-symbol used to indicate the beginning context

Probable combinations

atriptothethemepark $\rightarrow$ a trip to the the me park OR a trip to the theme park

N-gram model: A model that uses probabilities for a word, given its preceding word(s).

$N = 3$:

$\mathbf{P(a|PAD, PAD) * P(trip|PAD, a) * P(to|a, trip) * P(the|trip, to) * P(theme|to, the) * P(park|the, theme)}$

These kinds of probabilities are called: **conditional probabilities**

E.g., the probability of the word "*park*", when "*the theme*" are preceding words.

Conditional probability can, e.g., be computed through **maximum-likelihood estimate**: (Here for N=3)

$$P(w_i | w_{i-2}, w_{i-1}) = \frac{c(w_{i-2}, w_{i-1}, w_i)}{c(w_{i-2}, w_{i-1})}$$

Where c means the frequency in the training corpus for the n-gram.
E.g.: What is the probability of the word "park", given that the two previous words are "the theme".

$$P(\text{park} | \text{the, theme}) = \frac{c(\text{the, theme, park})}{c(\text{the, theme})}$$

*How many times does
"the theme park"
occur in the corpus?*

*How many times does
"the theme" occur in
the corpus?*

To create an n-gram language model, from a corpus: Store the frequencies of all n-grams of length n and $n-1$ that occur in your corpus.

E.g., if you have stored: " $c(\text{the, theme, park})$ " and " $c(\text{the, theme})$ ", you can use these values to calculate:

$$P(\text{park}|\text{the, theme}) = \frac{c(\text{the, theme, park})}{c(\text{the, theme})}$$

How many times does "the theme park" occur in the corpus?

How many times does "the theme" occur in the corpus?

Problem: (Especially likely for larger n.) What if, e.g., "to the theme" never occurs in your training corpus? Then you will divide by 0 when using your language model → undefined answer.

$$P(\text{park}|\text{to, the, theme}) = \frac{c(\text{to, the, theme, park})}{c(\text{to, the, theme})}$$

Solution: Add a small smoothing constant k to each n-gram frequency. (This can be viewed as pretending that each n-gram occurs k times more than it actually does.)

$$P(w_i|w_{i-2}, w_{i-1}) = \frac{k + c(w_{i-2}, w_{i-1}, w_i)}{kV + c(w_{i-2}, w_{i-1})}$$

V is total vocabulary size (= the total number of different words in the training corpus.)

After having collected the n-gram frequencies, we have the information needed to be able to compute the probability of a sentence.

$N = 3$:

”a trip to the theme park”

$\mathbf{P}(a|\text{PAD}, \text{PAD}) * \mathbf{P}(\text{trip}|\text{PAD}, a) * \mathbf{P}(\text{to}|a, \text{trip}) * \mathbf{P}(\text{the}|\text{trip}, \text{to}) * \\ \mathbf{P}(\text{theme}|\text{to}, \text{the}) * \mathbf{P}(\text{park}|\text{the}, \text{theme}) * \mathbf{P}(\text{END}|\text{theme}, \text{park})$

Use END as a last token.